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EXPT NO: 9

DATE: _____

A PYTHON PROGRAM TO PERFORM CRITICAL PATH METHOD (CPM) ANALYSIS

AIM:

To perform Critical Path Method (CPM) Analysis for project scheduling using Python, computing Early Start, Early Finish, Late Start, Late Finish, and Total Float for each activity.

PROCEDURE:

1. Define tasks with durations and predecessor relationships.
2. Perform forward pass to compute Early Start (ES) and Early Finish (EF) for each task.
3. Perform backward pass to compute Late Finish (LF) and Late Start (LS).
4. Compute Total Float = LS - ES for each activity.
5. Identify the Critical Path (activities with zero float) and display results.

PROBLEM 1: 6-Activity Project Network

CODE

```
def cpm_analysis(tasks, durations, predecessors):
    ES = {}; EF = {}
    for task in tasks:
        preds = predecessors[task]
        ES[task] = max((EF[p] for p in preds), default=0)
        EF[task] = ES[task] + durations[task]
    proj_dur = max(EF.values())
    LF = {}; LS = {}
    for task in reversed(tasks):
        succs = [t for t in tasks if task in predecessors[t]]
        LF[task] = min(LS[s] for s in succs) if succs else proj_dur
        LS[task] = LF[task] - durations[task]
    Float = {t: LS[t]-ES[t] for t in tasks}
    critical = [t for t in tasks if Float[t] == 0]
    print(f"{'Task':>4} {'ES':>4} {'EF':>4} {'LS':>4} {'LF':>4} {'Float':>6}")
    print("-"*32)
    for t in tasks:
        print(f"{t:>4} {ES[t]:>4} {EF[t]:>4} {LS[t]:>4} {LF[t]:>4} {Float[t]:>6}")
    print("Project Duration:", proj_dur)
    print("Critical Path:", " -> ".join(critical))
    tasks=['A','B','C','D','E','F']
    durations={'A':5,'B':3,'C':4,'D':6,'E':2,'F':3}
    predecessors={'A':[],'B':['A'],'C':['A'],'D':['B'],'E':['C'],'F':['D','E']}
    cpm_analysis(tasks, durations, predecessors)
```

OUTPUT

Task	ES	EF	LS	LF	Float

A	0	5	0	5	0
B	5	8	5	8	0
C	5	9	7	11	2
D	8	14	8	14	0
E	9	11	11	13	2
F	14	17	14	17	0
Project Duration: 17					
Critical Path: A -> B -> D -> F					

PROBLEM 2: 7-Activity Project Network

CODE

```
tasks=['A','B','C','D','E','F','G']
durations={'A':4,'B':6,'C':5,'D':3,'E':7,'F':4,'G':2}
predecessors={'A':[],'B':[],'C':['A'],'D':['A','B'],'E':['B'],'F':['C','D'],'G':['E','F']}
cpm_analysis(tasks, durations, predecessors)
```

OUTPUT

Task	ES	EF	LS	LF	Float

A	0	4	3	7	3
B	0	6	0	6	0
C	4	9	7	12	3
D	6	9	6	9	0
E	6	13	6	13	0
F	9	13	9	13	0
G	13	15	13	15	0
Project Duration: 15					
Critical Path: B -> D -> F -> G					

PROBLEM 3: 8-Activity Construction Project

CODE

```
tasks=['A','B','C','D','E','F','G','H']
durations={'A':3,'B':5,'C':4,'D':2,'E':6,'F':3,'G':4,'H':2}
predecessors={'A':[],'B':['A'],'C':['A'],'D':['B'],'E':['B','C'],'F':['D'],'G':['E'],'H':['F','G']}
cpm_analysis(tasks, durations, predecessors)
```

OUTPUT

Task	ES	EF	LS	LF	Float

A	0	3	0	3	0
B	3	8	3	8	0
C	3	7	4	8	1
D	8	10	10	12	2
E	8	14	8	14	0
F	10	13	12	15	2
G	14	18	14	18	0

```
H 18 20 18 20 0
```

```
Project Duration: 20
```

```
Critical Path: A -> B -> E -> G -> H
```

RESULT:

Thus, Critical Path Method (CPM) Analysis for Project Scheduling using Python has been implemented and executed successfully.
